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Export controls and innovation transfer within Chinese business groups: Evidence from the U.S. entity list

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ABSTRACT

This paper investigates the positive effect of U.S. export controls on the innovation performance of firms in Chinese business groups. Using annual microdata from 2010 to 2022, we analyze how firms indirectly affected by the U.S. entity list respond to these shocks. The positive effect of export controls on innovation performance is evident when indirectly shocked firms are peers of the blacklisted firms and becomes even more pronounced when they are upstream of the blacklisted firms, driven by the need for suppliers to initiate their own innovations. Additionally, we find that both capital and talent within the group are reallocated to these indirectly affected firms following the sanctions. These findings suggest that business groups serve as an internal innovation market, where innovation transfer occurs through inter-firm industrial interaction and resource reallocation. However, the shocks show no significant effect on the innovation performance of the group as a whole, rejecting the hypothesis of overall innovation enhancement. Further analysis reveals that export controls significantly improve firms' productivity, illustrating positive spillover effects on product markets. Overall, this study reveals that export controls drive innovation transfer within Chinese business groups and accelerate technological decoupling from the United States.

1. Introduction

Export controls, often implemented through legislative and administrative measures, are a common tool used to restrict the export of specific products or technologies to other countries for the purpose of national security or trade protectionism (Anwar et al., 2024). Particularly in the context of rising anti-globalization trends and escalating trade frictions, the deterioration of the Sino-US relations has sparked a trade war between the two nations (B. Liu et al., 2024b). Consequently, the United States has increasingly employed export controls to sanction Chinese companies, leaving them unable to access U.S. high-tech products and technologies.

A growing body of research has explored the impact of export controls on firms' innovation performance in sanctioned developing countries (Anwar et al., 2024; Huang et al., 2024; B. Liu et al., 2024b; X. Liu et al., 2024a). While most studies highlight the innovation-promoting effects of export controls by increasing the patented innovations of sanctioned firms, they often overlook the role of business groups, a distinctive form of business organization. In this paper, we focus on the pyramidal business groups, which are considered an organizational response to the weak institutional context of emerging economics, with important implications for firms and social welfare (Khanna and Palepu, 2000a). Specifically, we examine how U.S. export controls affect the innovation performance of firms in Chinese business groups and whether these groups function as internal innovation markets.

Identifying the causal effect of U.S. export controls on the innovation performance of firms in Chinese business groups is challenging. Since export controls primarily target innovative and high-tech firms to safeguard national security, studying their direct impact on sanctioned firms may introduce a strong endogeneity. To address potential reverse causality and other unobservable factors influencing both the likelihood of being sanctioned and innovation outcomes, we exploit the U.S. entity list as an exogenous shock. Firms in the same business group as blacklisted firms in and after the sanction year are treated as the treatment group, while firms in business groups with no entities ever blacklisted serve as the control group. Using a multiple-period difference-in-differences (DID) approach, we find that the U.S. entity list significantly enhances the innovation performance of firms in Chinese business groups. The result holds robust across various tests, including the inclusion of two-way fixed effects, controls for the potential impact of the

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0048-7333/© 2025 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

"Made in China 2025" policy, exclusion of samples affected by additional export tariffs, and alternative treatment group definitions and estimation methods.

We next explore the mechanisms through which export controls affects firms' innovation performance in business groups. First, we examine the internal transfer channel. Previous literature has suggested that business groups can create internal factor markets, reallocating resources among member firms to mitigate external market frictions and costs (He et al., 2013; Huneeus et al., 2021). This implies that business groups may serve as internal markets for innovation. More importantly, if business groups do function as internal innovation markets, innovation transfer may depend on the industrial interaction between sanctioned firms and other firms within the group. To investigate this, we categorize indirectly shocked firms into same-industry and different-industry groups, based on whether they are in the same industry as the blacklisted firms. Our results reveal that the positive effects of export controls are significant in same-industry groups. Due to the presence of information asymmetry, managerial agency problems and transaction costs, internal substitution is less costly than external substitution (He et al., 2013). Thus, export controls may induce a focus on peers of sanctioned firms in a business group, allowing them to continue innovating as substitutes, which we term peer innovation substitution. Further analysis shows no significant changes in technology acquisition among peers of blacklisted firms after sanctions, but a notable increase in invention patent purchases. This suggests that, in the short term, peer firms actively engage in patenting activities, effectively substituting for the innovation efforts of blacklisted firms.

On the other hand, our results indicate that the positive effects of export controls are more pronounced in different-industry groups. Within these groups, supplier-customer relationships emerge as critical channels influencing innovation. For upstream firms in the same business group as sanctioned firms, export controls may reshape customer demand structures by reducing reliance on foreign suppliers and increasing demand for domestic suppliers. This shift drives innovation among domestic suppliers, a mechanism we term supply chain-driven innovation progress. To examine this further, we categorize different-industry groups into supply-chain and non-supply chain groups. The results reveal that supply-chain relationships play a more important role in fostering innovation. Further, we find that in supply-chain groups, export controls on downstream firms push upstream firms to innovate independently, and accelerate their technological decoupling from the United States, verifying the mechanism of supply chain-driven innovation progress. Both peer competition-induced innovation substitution and supply chain-driven innovation progress underscore the existence of an internal market for knowledge and technology within business groups. We also find that both capital and talent are reallocated within the business groups following the shock. It demonstrates that export controls drive the reallocation of innovation resources within business groups from blacklisted firms to other firms, supporting the hypothesis of internal innovation transfer.

Second, we explore the overall innovation enhancement channel. Similar to individual firms, export controls could stimulate and enhance the innovation of business groups as a whole. By breaking dependence on imported production factors, export controls may prompt sanctioned groups to re-domesticate their supply chains and create additional market demand, thereby stimulating innovation and development in related industries. Furthermore, export controls may increase market competition pressure on directly impacted groups, encouraging greater R&D investment and innovative activities to enhance competitiveness. If export controls lead to an overall innovation boost for business groups, we would expect to observe improved innovation performance among indirectly shocked firms as well. To test this hypothesis, we aggregate the sample at the group level, treating groups with at least one firm included in the U.S. entity list as the treatment group, and groups where no firms are affected by the shock as the control group. We then re-run the baseline regressions. The results show no improvement in the innovation performance of the group as a whole, thus rejecting the overall innovation enhancement hypothesis.

Finally, to further assess the impact of export controls on productive activities in the real economy, we calculate the total factor productivity (TFP) of firms to examine the effect of the U.S. entity list on the productivity of firms within business groups. The results indicate that export controls significantly enhance firms' productivity, illustrating a positive spillover effect of export controls on product markets.

Our study contributes to three strands of literature. First, it enriches the literature on the consequences of international trade frictions by examining the impact of the U.S. entity list on firms in the Chinese business groups. There is no consensus on the impact of export controls on the innovation performance of firms in sanctioned countries. Firm-level studies on the impact of export controls are scarce (Anwar et al., 2024; Huang et al., 2024; B. Liu et al., 2024b; X. Liu et al., 2024a). Anwar et al. (2024) find that U.S. export controls stimulate technological innovation in Chinese sanctioned firms, while Huang et al. (2024) show that these controls significantly increase the number of patented innovations among Chinese targeted firms. However, to the best of our knowledge, no research has yet examined the impact of export controls on the innovation performance of firms within business groups, nor has any study explored the mechanisms through which export controls influence innovation from a technological perspective.

Second, our paper contributes to the literature on business groups by providing new evidences on the importance of the pyramidal business groups in the organizational structure of firms in developing countries (He et al., 2013). We demonstrate that business groups can create internal markets for innovation and reallocate innovation factors such as capital and talent in response to external shocks. More importantly, we examine how industrial interaction among firms, particularly supplier-customer relationships, affect innovation transfer within business groups. We extend the literature on supply-chain relationships (Chu et al., 2019) by highlighting the innovation resilience of supply-chains within business groups.

Third, this paper contributes to the broader literature on innovation and development in developing countries. As technological laggards and catch-up players, developing countries face both opportunities and challenges in fostering innovation and development. Our findings prove an unintended consequence of export sanctions imposed by developed countries: the promotion of technological innovation, and a degree of technological decoupling among firms in developing countries, which is consistent with the results of Han et al. (2024).

The rest of the paper is organized as follows. Section 2 provides background on the U.S. entity list and business groups in China. Section 3 develops the hypotheses. Section 4 describes the data and estimation strategy used in our model. Section 5 presents the empirical results. Section 6 discusses the mechanisms and provides further analyses. Section 7 concludes.

2. Background

2.1. The U.S. entity list

Export controls implemented by the United States comprise two main types of regulations: the International Traffic in Arms Regulations (ITAR) and the Export Administration Regulations (EAR). ITAR, regulated by the U.S. Department of State's Directorate of Defense Trade Controls, focuses on military products. In contrast, EAR, overseen by the Bureau of Industry and Security (BIS) of the U.S. Department of Commerce, governs dual-use products. EAR includes a series of lists such as the entity list, unverified list, denied persons list, and ultimate end user list, all of which are crucial for controlling the export of specific products and intangible technologies (Bown, 2019). This paper particularly focuses on the entity list, which is widely influential but underexplored.

Since its inception in February 1997, the entity list has become one of the most frequently used export control measures, encompassing a wide range of foreign entities, including enterprises, research institutions, and individuals. According to the entity list, any U.S. exporter is prohibited from exporting EAR-controlled products or technologies to listed entities without a license. Notably, the process for removing a foreign entity from the entity list involves a stringent and complex review by the End-User Review Committee (ERC) and requires unanimous approval, which makes removal a particularly challenging process.

Fig. 1 illustrates the global distribution of entities on the U.S. entity list, underscoring the widespread impact of U.S. export controls. By the end of 2022, the distribution of listed entities reveals a clear trend, with Russia, China, Pakistan, Iran, and the United Arab Emirates leading in the number of sanctioned entities. Among these, Huawei stands out as a noteworthy case. Since 2019, the U.S. Department of Commerce has imposed stringent sanctions on Huawei and its affiliates, adding over 150 associated companies worldwide to the entity list. These sanctions stipulate that any U.S. company exporting or re-exporting products to Huawei must adhere strictly to the Export Administration Regulations (EAR). As a result, Huawei faces substantial barriers in accessing U.S. products and technologies subject to the EAR. Moreover, items produced in other countries with 25 % or more U.S.-origin content are also restricted. The sanctions imposed on the entity list severely hinder Chinese firms like Huawei from importing high-tech products and absorbing knowledge and technology from the United States. The impact of these sanctions extends beyond technological giants like Huawei, affecting a wide range of Chinese firms. While primarily restricting technological exchanges and cooperation between China and the United States, these sanctions also drive Chinese entities to enhance their independent innovation efforts, seek alternative technologies and markets, and address the challenges posed by external sanctions.

2.2. Business groups in China

A business group is defined as a collection of legally independent firms bound together by persistent formal and informal ties (Granovetter, 2010; Khanna and Rivkin, 2001; Khanna and Yafeh, 2007; Morck et al., 2005). A key feature of a business group is that its firms are ultimately controlled by the same shareholder, which is often referred to as a complex pyramidal structure (Huneeus et al., 2021). In developing countries, more than half of all listed firms are affiliated with business groups, and a significant portion of firms in developed countries are also part of business groups (Huneeus et al., 2021; Khanna and Yafeh, 2007; Morck et al., 2005).

Chinese business groups emerged China's market-oriented reforms, when the Chinese government sought to foster business groups to absorb new technologies, improve financial performance, and enhance international competitiveness (He et al., 2013). With the continued progress of reform and opening up, the reform of state-owned enterprise management began in the 1980s, leading to the formation of cross-regional and cross-industry economic joint enterprise groups, representing the early stages of modern business groups (Li and Yan, 2021; Cai et al., 2023). To prevent the dispersed property rights of state-owned enterprises from hindering the development of business groups, the Chinese government proposed the creation of a specialized state-owned asset management agency to serve as the state's representative funder (Cai et al., 2019). The most direct manifestation was the establishment of the State-owned Assets Supervision and Administration Commission (SASAC) in 2003, which centralized the funding and oversight of state-owned enterprises' property rights. In response, local governments created a number of sizable business groups to align with the central government's call for a strategic reorganization of the state-owned economy. Until the central government explicitly proposed in the "12th Five-Year Plan" that "the cultivation of a number of large-scale business groups with international competitiveness will be the focus of enterprise reform and development in the future", China's business groups began to enter the fast lane of high-speed development (Cai et al., 2023).

As of the end of 2022, there were 429 business groups of various types in A-share listed companies on the Shanghai and Shenzhen Stock Exchanges, involving 1520 member listed companies. These account for 29.53 % of the total number of A-share listed companies during the same period, according to our sample.[1] Consistent with the statistics in the literature (Cai et al., 2023; He et al., 2013; Yiu et al., 2005), our findings show that, although the number of group listed companies accounts for less than one-third of all listed companies, they have accounted for nearly 70 % of total market capitalization of A-share listed companies. It implies that group-organized firms have become a significant part of China's capital market. Furthermore, their complex shareholding structures confer a unique role in corporate operations and governance, distinguishing them from individual firms.

3. Hypotheses

Previous studies have explored the impact of export controls on innovation from the perspectives of both policy-enforcing and sanctioned countries (Hosoe, 2021; Marukawa, 2013). However, there is no consensus regarding the effect of export controls on the innovation performance of firms in sanctioned countries. The related literature can be categorized into two main perspectives: negative effects and positive effects, which we discuss in detail below.

3.1. Negative effects of export controls on sanctioned firms' innovation performance

A strand of literature suggests that export control policies negatively impact the innovation performance of sanctioned countries. Krugman (1979) develops a general-equilibrium model of product cycle trade, illustrating the motivations behind protectionism and export controls: while technology transfers from the innovating North to the non-innovating South through trade, export controls can prevent the transfer and diffusion of technology to the laggard, safeguarding the technological lead and interests of the advanced. As imports are important in promoting innovation in technology-lagging countries, export controls can hinder knowledge spillovers and follow-on innovation in sanctioned countries (Anwar et al., 2024). Moreover, Export controls can disrupt supply chains, reduce overall productivity, and lead to significant welfare losses, all of which impede local firms' research and development efforts (Hosoe, 2021). Some scholars have reached similar results through real cases. Choi and Niculescu (2006) find that the U.S. export controls have negative impacts on the Canadian space industry, particularly with respect to costs and timelines, even though non-US technology is comparable or even superior to the American equivalent in most cases. Similarly, Marukawa (2013) contends that exporting high-tech items and transferring technology from Japan to China may advance China's technological capabilities while eroding Japan's competitive edge. Hence, any tightening of controls on high-tech exports from Japan may be detrimental for Chinese industries. Aforementioned studies explain the potential negative impact of export controls from the perspective of trade costs, productivity, knowledge spillovers and technology transfer.

3.2. Positive effects of export controls on sanctioned firms' innovation performance

Another strand of literature, increasingly acknowledged, highlights the positive effects of export controls on the innovation performance of sanctioned firms. Anwar et al. (2024) find that U.S. export controls spur technological innovation in Chinese sanctioned firms by increasing government grants, boosting R&D investment, and prompting firms to

[1] A detailed definition of a business group is provided in the empirical design section of this paper.

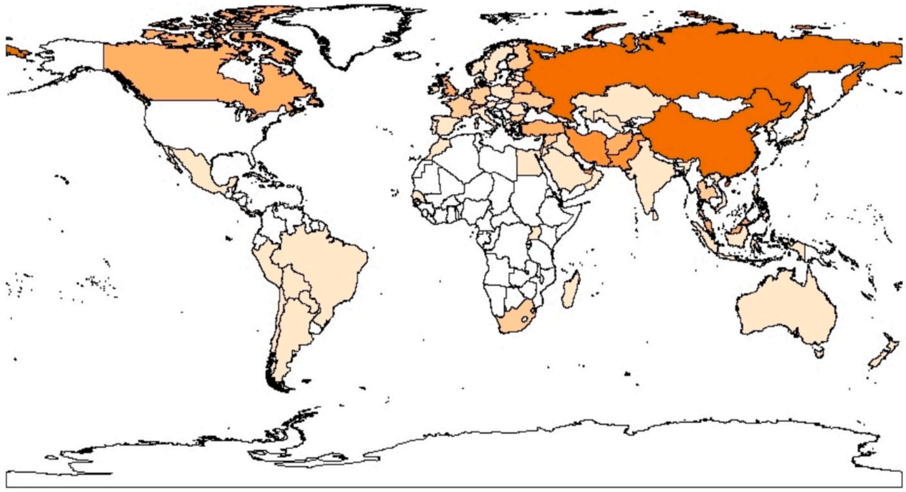

Fig. 1. Number of entities sanctioned by the U.S entity list in each country. Notes: This figure illustrates the global distribution of entities on the U.S. entity list by the end of 2022. Orange areas represent the regions where the sanctioned entities are located, with darker shades indicating a higher concentration of sanctioned entities. White areas represent regions with no sanctioned entities.

adjust their business operations. Huang et al. (2024) find that U.S. export controls significantly increase the number of patented innovations of Chinese targeted firms. And targeted firms increase more-substantive innovations that increase the value-added of their products, indicating export controls reduce reliance on foreign technology and encourage innovative activities to improve productivity and competitiveness. Similarly, B. Liu et al. (2024b) indicate that the inclusion of Chinese firms in the entity list disrupts their dependency on foreign production inputs and motivates them to enhance technological M&As. X. Liu et al. (2024a) demonstrate that while U.S. export controls reduce profits and productivity of sanctioned Chinese firms, these firms are more likely to maintain positive R&D expenditures and hold patents.

While these studies highlight the positive effects of export controls on innovation, few delve into the underlying mechanisms in the context of U.S.-China technological decoupling and China's drive for independent innovation. To the best of our knowledge, no prior research has yet examined the impact and mechanisms of export controls on the innovation performance of firms in business groups in developing countries. As mentioned in Section 2.2, business groups, due to their complex pyramidal ownership structures, play a unique role in corporate operations and governance compared to individual firms. So can business groups serve as internal innovation markets, facilitating the allocation of innovation factors such as knowledge and technology among their member firms? How does the interaction among economically interlinked member firms, especially industrial interaction, influence the impact of export controls on innovation performance of firms in business groups? We address these questions by examining two key mechanisms through which export controls exert positive effects: internal innovation transfer and overall innovation enhancement.

3.2.1. Internal innovation transfer

(a) Interaction among firms

If business groups create an internal market for innovation, interaction among firms, especially industrial interaction closely tied to innovation activities, would influence the impact of export controls on firms' innovation performance. Relevant literature generally classifies the industrial relations into horizonal relations (peer substitution) and vertical relations (supply chain complementarity).

With respect to horizonal relationships, peer firms often serve as less costly substitutes compared to external market alternatives. This advantage stems from reduced information asymmetry, fewer agency problems, and lower transaction costs associated with intra-group coordination relative to arm's-length market exchanges (He et al., 2013). The effect is particularly pronounced in emerging markets such as China, where external markets are often underdeveloped and formal institutional protections remain weak (Ghemawat and Khanna, 1998; Khanna and Yafeh, 2007). In such contexts, business groups tend to rely more heavily on internal markets to circumvent external market inefficiencies and to reduce transaction frictions and costs (Chang and Choi, 1988; Khanna and Palepu, 2000a; Khanna and Palepu, 2000b; Fisman and Wang, 2010). Almeida and Wolfenzon (2006) further argue that pyramidal ownership structures facilitate strategic redeployment across affiliated firms, strengthening their ability to substitute for one another in operational functions. Thus, for firms within a business group, export controls on one firm may induce a focus on other firms in the same industry, prompting these firms to pursue innovation as substitutes, which we term peer innovation substitution. We propose the following internal market hypothesis:

Hypothesis 1. (H1): Export controls lead to peer innovation substitution within business groups, transferring innovation efforts from blacklisted firms to other group firms.

For vertical relationships, a growing body of literature has examined the effects of supply-chain relationships on corporate investment decisions. However, technological innovation, as a special type of corporate investment, is rarely explored. A few theoretical models on innovation suggest that innovation enhances the production process by reducing marginal costs (Kamien et al., 1992; Leahy and Neary, 1997). Thus, a supplier's incentive to innovate is closely linked to the quantity of products (services) it produces (provides) for its customer (Chu et al., 2019). Besides, suppliers and customers often share important production factors, such as intermediate inputs, talent, and natural resources (Orlando, 2004), which play a pivotal role in fostering suppliers' innovation. There are other explanations of the impact of supply-chain relationships on firms' innovation as well. For example, Dasgupta et al., 2019 document that a close social connection between managers of suppliers and customers helps mitigate hold-up problems and enhance supplier innovation.

In our study, export controls may alter the demand structure of customer firms by reducing reliance on foreign suppliers while increasing demand for domestic suppliers. It will drive domestic suppliers to innovate and accelerate technological decoupling from the United States (Han et al., 2024). Lower communication costs, stronger social connections, and greater factor-sharing with domestic suppliers further incentivize upstream firms to innovate. Therefore, we propose the following internal market hypothesis:

Hypothesis 2. (H2): Export controls lead to supply chain-driven innovation progress within business groups, shifting innovation efforts from blacklisted firms to other group firms.

(b) Resource allocation of business groups

Previous literature offers several explanations for the complex pyramidal ownership structure of business groups. One prominent view is that business groups create internal factor markets to alleviate external market frictions and reallocate resources among member firms. For instance, business groups can function as internal financial markets, facilitating capital allocation among firms (Bena and Ortiz-Molina, 2013; He et al., 2013; Khanna and Palepu, 2000a). Khanna and Palepu (2000a) find that internal capital markets within Indian groups operate effectively like financial markets in developed countries. Bena and Ortiz-Molina (2013) find that European business groups offer financing advantages in establishing new firms when the pledgeability of cash flows to outside funders is constrained. He et al. (2013) provide evidence supporting the role of business groups in risk sharing among affiliated firms in China. Especially in countries with weak investor protection or high external capital costs, a business group may have extensive governance functions to mitigate the financial constraints faced by firms and ensure close monitoring of management decisions. Therefore, we argue that business groups can reallocate capital from impacted firms to unaffected firms after the implementation of export controls. We propose the following internal market hypothesis:

Hypothesis 3. (H3): Export controls drive the reallocation of capital within business groups, shifting resources from blacklisted firms to other group firms.

Besides, business groups can serve as internal labor markets, facilitating the flow of workers among firms (Faccio and O'Brien, 2021; Huneeus et al., 2021). Faccio and O'Brien (2021) find that group-affiliated firms exhibit smaller employment fluctuations compared to unaffiliated firms during macroeconomic shocks including booms and recessions, supporting the explanation of internal labor markets. Huneeus et al. (2021) provide evidence of labor mobility inside business groups, indicating that joint ownership eases the redeployment of workers endowed with general management skills. They also show that the reallocation of top talent within groups is more sensitive to international shocks. Based on the above evidence, we propose the following internal market hypothesis:

Hypothesis 4. (H4): Export controls drive the reallocation of talent within business groups, transferring skilled workers from blacklisted firms to other group firms.

3.2.2. Overall innovation enhancement

Apart from the explanation that business groups act as internal innovation markets, where internal innovation transfer occurs through inter-firm interaction and resource reallocation, another perspective is that export controls may positively influence the overall innovation performance of business groups. Like individual firms, export controls can stimulate innovation in business groups by reducing reliance on imported production inputs, encouraging the re-domestication of supply chains, and creating additional market demand that fosters innovation in related industries. Export controls may also intensify market competition for directly impacted groups, promoting group R&D investment and innovative activities to enhance competitiveness. Hence, we propose the overall innovation enhancement hypothesis for business groups:

Hypothesis 5. (H5): Export controls improve the innovation performance of business groups as a whole.

4. Data and estimation strategy

4.1. Data and variables

We construct a sample of Chinese A-share listed firms on the Shanghai and Shenzhen Stock Exchanges from 2010 to 2022. The main datasets used in this paper are obtained from several sources. First, we collect detailed information on the ultimate controlling shareholders and ownership structure of China's listed companies from their annual reports which are collated by the China Stock Market and Accounting Research (CSMAR) database. To ensure accuracy, we manually verify and correct it with the ownership structure charts. Following He et al. (2013), we identify two or more listed firms belonging to the same business group if they share the same ultimate controlling shareholders. Second, we manually obtain the blacklisted Chinese listed firms and the corresponding years of inclusion in the U.S. entity list, published by the U.S. Bureau of Industry and Security. We translate the Chinese entities' English names into Chinese, and check them with the Qichacha platform[2] based on their names and registered addresses. Through manual searches on Qichacha, we identify whether the entity is a listed company, a subsidiary, or a sub-subsidiary. These entities are then matched with the database of Chinese listed companies to identify blacklisted firms. Notably, in the baseline regression, we treat listed companies themselves or their subsidiaries or sub-subsidiaries sanctioned as blacklisted firms. But in robustness tests, we only consider listed companies themselves included in the entity list as blacklisted firms. Third, data on patent applications and invention patent applications are sourced from the incoPat database, which currently covers over 180 million patent documents from 170 countries, organizations and regions worldwide. This database contains nearly 200 variables, and includes detailed information on patents, laws, operations, cognates, citations, etc., which can meet the needs of this study. Finally, control variables for firms are obtained from the CSMAR database.

The dependent variables used to measure firms' innovation performance are the number of patent applications (*lnpatent*) and invention patent applications (*lninvent*). As previous literature has demonstrated, innovations are usually officially introduced to the public in the form of patents with detailed information, making patents the most important indicator of innovation output (Griliches, 1990; Gao and Chou, 2015). So we take the natural logarithm of one plus the number of patent applications and invention patent applications to mitigate the right skewness of the dependent variables. Besides, we construct two other variables, *efficiency* and *efficiency_inv* to measure innovation efficiency, which captures innovation output per unit of input. Innovation input is measured by the firm's annual R&D expenditure, whereas innovation outputs are measured by patent applications and invention patent applications, respectively. Both patents and efficiency provide a measurable and less biased assessment of firms' R&D productivity.

The independent variable is the U.S. entity list status (*entitylist*), a dummy variable that indicates whether or not firms in Chinese business groups are included in U.S. entity list in a given year. Since the majority of blacklisted firms are technology-based, and one characteristic of such firms is their innovation and frequent patent applications, directly studying the impact of the U.S. entity list on these firms could lead to a strong endogeneity. To better identify the causal relationship, we use other firms in the same business group that are indirectly affected by the shock, as the treatment group. Firms in business groups that have never been included in the U.S. entity list are treated as the control group. Fig. 2 depicts our selection rules for treatment and control groups. We take *entitylist* as a value of 1 if the firm is in the treatment group and is in the year following its inclusion in the U.S. entity list, 0 otherwise. Notably, firms are added to the U.S. entity list in different years.

Following related literature, we also include firms' fundamental

[2] Website: https://www.qcc.com.

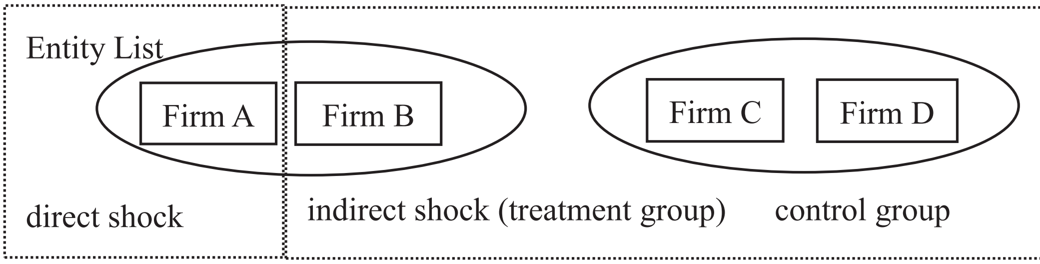

Fig. 2. The setting of the independent variable. Notes: Fig. 2 illustrates the selection rules for treatment and control groups: Firm A represents firms included in the entity list, which are directly shocked; Firm B (treatment group) represents other firms within the same business group as Firm A, but not directly shocked; Firm C and Firm D (control group) are firms that have never been sanctioned.

attributes, financial and operational capabilities, and governance capabilities as control variables, including corporate size (*size*), corporate age (*age*), corporate leverage (*lev*), cash holdings (*cash*), return on assets (*roa*), growth capacity of operating income (*growth*), CEO duality (*dual*), shareholding ratio of the largest shareholder (*top*1), and independent director proportion (*indep*). *size* and *age* are fundamental attributes that we control for to capture the impact of resource accumulation and experience on shaping firms' innovation capabilities (Anwar et al., 2024; Balasubramanian and Lee, 2008). *lev*, *cash*, *roa* and *growth* control for financial and operational capabilities (B. Liu et al., 2024b). We also control for governance capabilities using *dual*, *top*1 and *indep* (B. Liu et al., 2024b; Lyu et al., 2024). Detailed variable definitions are presented in Table 1.

In addition, firms included in the sample must meet the following conditions: (1) The ultimate controller of the firm must control at least two member listed firms in the year, in order to construct a business group. (2) We focus on firms with normal listing status, so firms flagged with ST or *ST are excluded.[3] (3) Firms in the financial sector generally have less substantive innovation activities, so these firms are also excluded from the sample. The final sample contains 11,131 firm-year observations for 1520 Chinese listed firms from 2010 to 2022. Table 2 presents the descriptive statistics for the main variables used in this study. The variables in the sample reflect good variation.

Table 1
Variable definitions.

Variable name	Variable description
Dependent variable	
lnpatent	The number of patent applications of firms, equals the natural logarithm of one plus patent counts.
lninvent	The number of invention patent applications of firms, equals the natural logarithm of one plus invention patent counts.
efficiency	Innovation efficiency to capture innovation output per unit of input, equals patent applications divided by R&D expenditure.
efficiency_inv	Innovation efficiency to capture innovation output per unit of input, equals invention patent applications divided by R&D expenditure.
Independent variables	
entitylist	Dummy variable, equals one when a firm is in a business group with blacklisted firms following the year being sanctioned by the U.S. entity list, and zero otherwise.
Control variables	
size	Corporate size, equals the natural logarithm of the book value of total assets.
age	Corporate age, equals the natural logarithm of the difference between year t and the year the firm was listed.
lev	Corporate leverage, equals corporate total liabilities divided by the book value of total assets.
cash	Cash holdings, equals cash and cash equivalents divided by the book value of total assets.
roa	Return on assets, equals corporate net profit divided by the book value of total assets.
growth	Corporation growth capability, equals the growth rate of operating income.
dual	Dummy variable, equals 1 if chairman and general manager is the same person and 0 otherwise.
*top*1	Shareholding concentration, equals the share proportion of the largest shareholder.
indep	Independent director proportion, equals the proportion of independent directors to the number of directors on the board.

Table 2
Descriptive statistics.

Variables	Obs	Mean	SD	Min	Median	Max
lnpatent	11,131	1.711	1.915	0	1.099	6.868
lninvent	11,131	1.263	1.616	0	0	6.125
efficiency	9420	0.004	0.011	0	0.001	0.081
efficiency_inv	9420	0.002	0.004	0	0	0.031
entitylist	11,131	0.124	0.329	0	0	1
size	11,131	22.897	1.462	20.061	22.753	26.974
age	11,131	2.586	0.708	0	2.773	3.367
lev	11,131	0.504	0.205	0.075	0.511	0.937
cash	11,131	0.178	0.126	0.014	0.146	0.620
roa	11,131	0.031	0.053	−0.190	0.030	0.182
growth	11,131	0.449	1.345	−0.689	0.120	10.163
dual	11,131	0.111	0.314	0	0	1
top1	11,131	39.352	15.320	11.390	38.330	76.440
indep	11,131	0.314	0.064	0.182	0.313	0.500

Notes: Table 2 presents descriptive statistics for the main variables used in the study. The sample contains 11,131 firm-year observations for 1520 Chinese listed firms from 2010 to 2022, with some missing values for *efficiency* and *efficiency_inv*. We take the natural logarithm of all continuous variables and winsorize all continuous variables at the 1 % and 99 % levels to mitigate the impact of extreme values.

4.2. Empirical estimation

To examine the impact of export controls on the innovation performance of firms and investigate the internal innovation transfer responses within Chinese business groups, we treat the U.S. entity list as an exogenous shock and employ a multiple-period difference-in-differences (DID) approach. The empirical specification is as follows:

$$Y_{i,t} = \beta_0 + \beta_1 entitylist_{i,t} + X_{i,t}\gamma + firm_i + year_t + \varepsilon_{i,t} \tag{1}$$

where the subscripts *i* and *t* denote the firm and year, respectively. $Y_{i,t}$ measures the outcome variables of interest (e.g., *lnpatent*, *lninvent*, *efficiency* and *efficiency_inv*) for firm *i* in year *t*. $entitylist_{i,t}$ is a dummy variable indicating whether firm *i* is indirectly shocked by the U.S. entity list in year *t*. To isolate the effect of entity list, we also include $X_{i,t}$, a

[3] ST and *ST are abbreviations for 'Special Treatment', indicating that a company has sustained continuous losses or faced other significant financial risks, and has been given a risk warning by the stock exchange. The latter, *ST, represents a more severe condition than ST.

vector of time-varying firm-level control variables that potentially affect firms' innovation performance, as outlined in Section 3.1. $firm_i$ represents firm fixed effects, which capture unobservable time-invariant characteristics of firms. $year_t$ denotes year fixed effects that control for aggregate time-varying shocks or trend common to all firms, such as annual fluctuations or macro policies affecting innovation. $\varepsilon_{i,t}$ represents the random error term of the regression model. Our main interest in Eq. (1) is the coefficient β_1, capturing the average difference in innovation performance between firms indirectly affected by the U.S. entity list and those unaffected firms in business groups. If the coefficient is significantly positive, it means that firms indirectly affected by the U.S. entity list in business groups experienced an increase in innovation performance, otherwise, firms' innovation performance is either unaffected or negatively impacted.

Furthermore, a necessary identifying assumption for the DID estimation is that the treatment and control groups maintain similar trends before the shock occurs, i.e., the parallel trends assumption must be satisfied. To test this assumption, we conduct an event study to examine the parallel trend and explore the dynamic effects of the shock:

$$Y_{i,\ t} = \beta_0 + \sum_{k=-10,k\neq-1}^{6} \beta_k entitylist_{i,t+k} + X_{i,t}\gamma + firm_i + year_t + \varepsilon_{i,t} \tag{2}$$

where $t+k$ represents the relative period when the blacklisted firms are added to the entity list. The dummy variables for non-blacklisted firms are all set to zero. To mitigate the impact of outliers, we winsorize the period at the 1 % and 99 % levels (Beck et al., 2010). In other words, β_{-10} equals one for all years that are 10 or more year before the blacklisted firm is added to the entity list, while β_6 equals one for all years that are 6 or more year after the blacklisted firm is added to the entity list. We also drop the dummy variable $entitylist_{i,t-1}$ to avoid the multicollinearity issues. It ensures that the post-treatment effects are referenced relative to the period before firms were added to the entity list (Berniell, 2021; Chen and Lan, 2020). The remaining variables are kept consistent with Eq. (1) above.

5. Results

5.1. Baseline results

In Table 3, we examine the impact of the U.S. entity list on firms' innovation performance by estimating Eq. (1). In Column (1) and (3), the dependent variable is the number of patent applications, while in Column (2) and (4), the dependent variable is the number of invention patent applications. In the first two columns, we control for firm and year fixed effects only, without including any control variables. In Column (3) and (4), we further control for firm-level control variables. The results in all four columns show that the coefficients for the effect of the indirectly entity list shock on firms in business groups, *entitylist*, are significantly positive at the 1 % level. In term of magnitude, the number of patents for indirectly affected firms increases by 11.65 percentage points on average compared to unaffected firms after sanctions (Column (3)), while the number of invention patents rises by an average of 19.62 percentage points (Column (4)).

In Column (5) and (6), we replace the dependent variable with *efficiency* and *efficiency_inv* for robustness checks, and the results remain significantly positive. It implies that entity list sanctions also improve indirectly affected firms' innovation efficiency compared to unaffected firms in business groups.

Across all specifications, our baseline results consistently demonstrate that export controls have a significantly positive and causal impact on Chinese firms' innovation performance in business groups.

Fig. 3 illustrates the coefficients from the event study analysis by estimating Eq. (2), along with 95 % confidence intervals. The results show that the coefficients of the year dummy variables are all around zero and insignificant before the implementation of the entity list, confirming no significant differences in innovation performance between the treatment and control groups prior to the export controls. It satisfies the parallel trend assumption of the DID analysis, validating the control groups as suitable counterfactuals for the treatment groups. After sanctions, the coefficient estimates of *entitylist* become significantly positive and exhibit upward trends. It is worth mentioning that patent applications and invention patent applications show a fast growth response after the shock occurs, while innovation efficiency and invention innovation efficiency start to grow significantly in the third and fourth years after the shock (t=3, 4). According to Cruz-Cázares et al. (2013), improvements in innovation efficiency involve a more complex process, which requires more time to upgrade. These findings reinforce the robustness of the baseline results, indicating that export controls do have a positive effect on Chinese firms' innovation performance in business groups.

5.2. Robustness checks

In this section, we perform a series of robustness checks to validate our results further. First, regional-level factors, such as local subsidy policies and innovation incentives, may act as omitted variables that influence the baseline results. To address this concern, we add city-year fixed effects to control for time-varying unobservable biases at the city level. The results, reported in Panel A of Table 4, show that the coefficients of *entitylist* remain significantly positive. Similarly, considering the potential impact of industry-specific incentive policies, we include industry-year fixed effects, and account for both city-year and industry-year fixed effects (shown in Appendix Table A1 and Table A2). The estimates are consistent with the baseline results, indicating the robustness and reliability of our findings even after controlling for regional and industry-specific time-varying factors.

Second, domestic industrial policies during the same period, such as the “Made in China 2025” initiative, may have an impact on technological advancements, potentially biasing the baseline results. Announced in 2015, this policy aims to transform China into an innovation-driven manufacturing powerhouse (Li and Branstetter, 2024). We hypothesize that “Made in China 2025” positively affects firms' innovation activities. To disentangle the effects of the U.S entity list and “Made in China 2025”, we control for *made*2025, a variable equal to 1 if the firm is in the treatment group after 2015, and 0 otherwise. Panel B of Table 4 presents the results, showing that the coefficients for *entitylist* are still significantly positive, while those for *made*2025 are insignificant, indicating no mixed policy effects from “Made in China 2025”.

Similarly, the widespread tariff increases on Chinese exports during the U.S.-China trade war since 2018 may have a potentially huge impact on firms' production, operations and innovation. To exclude the potential effects of these additional export tariffs on firm innovation, we remove the sample affected by the tariffs. Panel C of Table 4 presents the results, which remain robust. Notably, all coefficients become larger, suggesting that the additional U.S. tariffs may have led to an underestimation of our baseline results.

Third, in the baseline regression, we also treat listed companies whose subsidiaries or sub-subsidiaries are sanctioned as blacklisted firms. Considering the potential differences in impact between sanctions targeting the list company itself and its subsidiaries or sub-subsidiaries, we remove this part of samples and treat only listed companies sanctioned of their own as blacklisted firms. The re-estimated baseline regression results, shown in Panel D of Table 4, confirm the robustness of our findings.

Fourth, considering the potential problems of natural logarithm transformation, we replace the dependent variables with the original number of patent applications and invention patent applications, and run the regression again using zero-inflated Poisson and zero-inflated Negative Binomial methods. The results are shown in Appendix

Table 3
Baseline results.

VARIABLES	(1) *lnpatent*	(2) *lninvent*	(3) *lnpatent*	(4) *lninvent*	(5) *efficiency*	(6) *efficiency_inv*
entitylist	0.1331***	0.2059***	0.1165***	0.1962***	0.0010**	0.0004***
	(0.043)	(0.038)	(0.043)	(0.038)	(0.000)	(0.000)
size			−0.0356	−0.0061	−0.0017***	−0.0007***
			(0.032)	(0.025)	(0.000)	(0.000)
age			0.0637	0.0910**	0.0002	0.0003*
			(0.045)	(0.039)	(0.000)	(0.000)
lev			0.0180	0.1031	0.0018	0.0009*
			(0.117)	(0.097)	(0.001)	(0.001)
cash			−0.2193*	−0.1430	−0.0011	0.0001
			(0.122)	(0.104)	(0.001)	(0.000)
roa			−0.0843	0.2755	−0.0037	−0.0001
			(0.236)	(0.199)	(0.003)	(0.001)
growth			0.0085	0.0026	−0.0000	−0.0000
			(0.006)	(0.005)	(0.000)	(0.000)
dual			0.0613*	0.0107	0.0008*	0.0003*
			(0.034)	(0.029)	(0.000)	(0.000)
top1			−0.0097***	−0.0068***	−0.0000**	−0.0000**
			(0.002)	(0.001)	(0.000)	(0.000)
indep			0.1672	0.0915	−0.0016	−0.0009
			(0.168)	(0.143)	(0.002)	(0.001)
Constant	1.6921***	1.2359***	2.6972***	1.3439**	0.0437***	0.0164***
	(0.010)	(0.008)	(0.721)	(0.568)	(0.007)	(0.003)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,989	10,989	10,989	10,989	9273	9273
R-squared	0.828	0.820	0.829	0.821	0.450	0.452

Notes: Based on firm-year-level regressions for the sample period of 2010 to 2022, we examine the impact of the U.S. entity list on firms' innovation performance in this table. All variables are defined in Table 1. All regressions include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

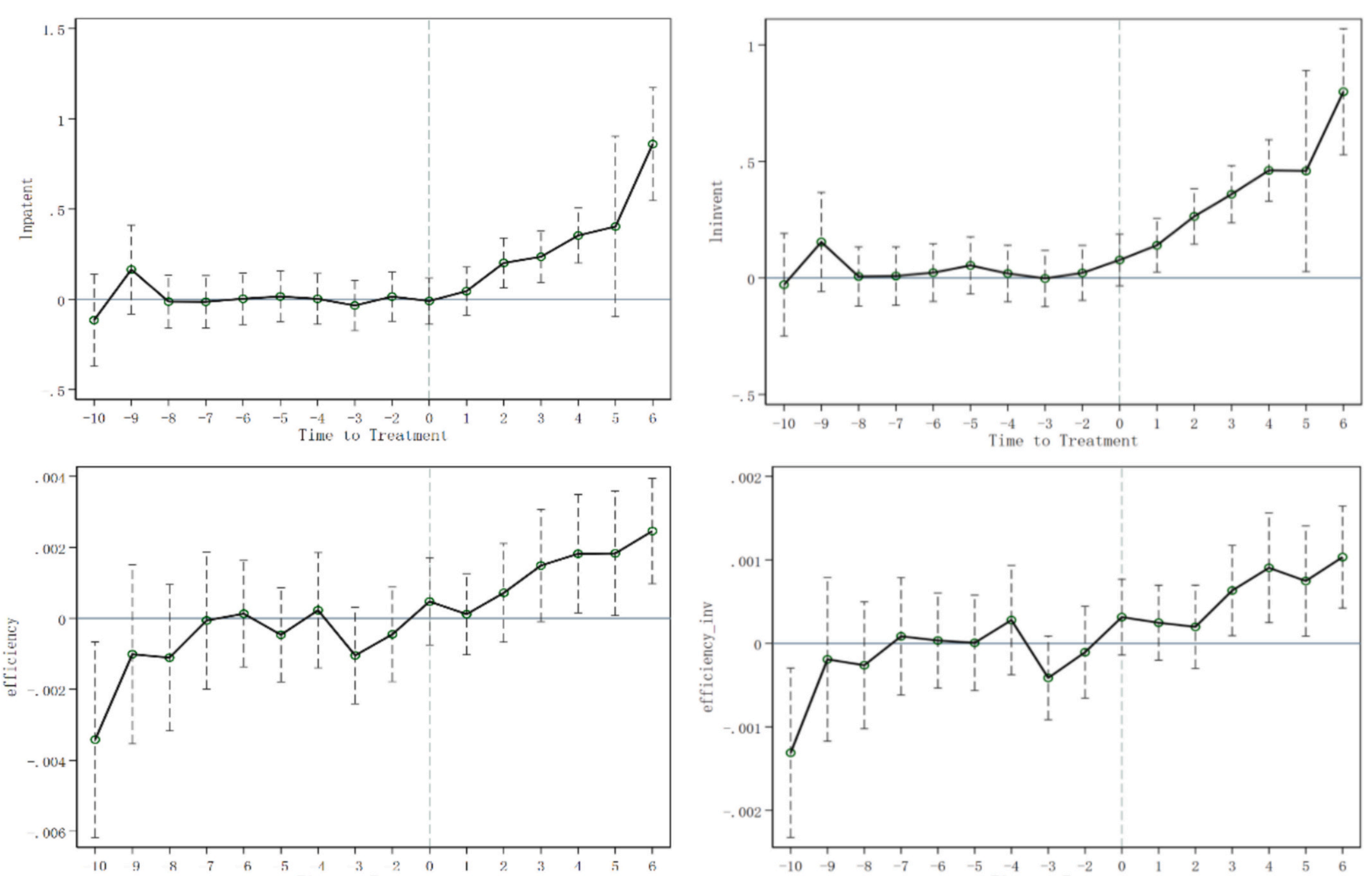

Fig. 3. The results of the parallel trend test. Notes: Fig. 3 shows the results of the parallel trend test. The first row presents the test results for *lnpatent* and *lninvent*, respectively. The second row shows the test results for *efficiency* and *efficiency_inv*, respectively.

Table 4
Robustness tests.

	(1)	(2)	(3)	(4)
VARIABLES	*lnpatent*	*lninvent*	*efficiency*	*efficiency_inv*
Panel A: DID estimate, including city-year fixed effects				
entitylist	0.1203**	0.2412***	0.0007	0.0004**
	(0.051)	(0.045)	(0.000)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes
Observations	9783	9783	8087	8087
R-squared	0.859	0.852	0.546	0.560
Panel B: DID estimate, controlling the impact of "Made in China 2025"				
entitylist	0.1120**	0.1927***	0.0009**	0.0004**
	(0.044)	(0.039)	(0.000)	(0.000)
made2025	−0.0392	−0.0307	−0.0009	−0.0004
	(0.066)	(0.057)	(0.001)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	10,989	10,989	9273	9273
R-squared	0.829	0.821	0.450	0.452
Panel C: DID estimate, excluding samples affected by trade wars				
entitylist	0.1871***	0.2678***	0.0019**	0.0009***
	(0.055)	(0.048)	(0.001)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	6438	6438	5035	5035
R-squared	0.815	0.803	0.455	0.457
Panel D: DID estimate, replacing the sample of blacklisted firms				
entitylist	0.1259***	0.2100***	0.0011**	0.0005***
	(0.044)	(0.038)	(0.000)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	10,989	10,989	9273	9273
R-squared	0.829	0.821	0.450	0.452
Panel E: DID estimate with cross-sectional nearest neighbor propensity score matching				
entitylist	0.1147***	0.1950***	0.0010**	0.0005***
	(0.043)	(0.038)	(0.000)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	10,904	10,904	9191	9191
R-squared	0.828	0.819	0.449	0.453
Panel F: DID estimate, controlling for direct government fiscal subsidies				
entitylist	0.1178***	0.1968***	0.0010**	0.0004***
	(0.043)	(0.038)	(0.000)	(0.000)
lnsubsidy	0.0087***	0.0038	0.0001**	0.0000*
	(0.003)	(0.003)	(0.000)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	10,989	10,989	9273	9273
R-squared	0.829	0.821	0.450	0.452

Notes: This table presents the results of robustness tests. All variables are defined in Table 1. All regressions control for firm-level control variables and include fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table A3. The coefficients for *entitylist* remain significantly positive, confirming the robustness of our results. Additionally, according to Damioli and Marin (2024), we also use the inverse hyperbolic sine transformation. The results (Appendix Table A4) are still robust.

Fifth, changes in the shareholding membership of a business group can affect identification, even though the range of shareholdings is stable for most of the sample we use. Therefore, we manually remove samples with changes in shareholdings and the results are shown in Appendix Table A5. The results are consistent with the baseline results, indicating that our identification is plausible. Additionally, to address the bias caused by the different treatment timing of the multiple-period difference-in difference (DID) approach, we use the estimation methods proposed by Borusyak et al. (2024), Callaway and Sant'Anna (2021), Sun and Abraham (2021), and Cengiz et al. (2019), respectively. The results are shown in Figs. 4 and 5, indicating that our baseline results are robust.

Sixth, we adopt the propensity score matching (PSM) approach to address potential endogeneity caused by selection bias. Specifically, we apply two methods: treating the sample as cross-sectional data for PSM (shown in Panel E of Table 4), and conducting PSM year-by-year, then aggregating the results into panel data (shown in Appendix Table A6). We utilize the nearest neighbor matching for both, with matching variables consistent with the baseline regression controls.[4] The results are still robust across both approaches.

Seventh, we control for direct government fiscal subsidies to address time-varying firm-level changes in innovation policies and incentives (shown in Panel F of Table 4). *lnsubsidy* is the natural logarithm of the fiscal subsidies. The results are robust. Additionally, we also control for typical policies of indirect tax credits, the InnoCom program and additional deduction from R&D expense. Appendix Table A9 and Table A10 indicate that the results remain robust.

Eighth, to account for the possibility that some applications may be denied during the examination process, we use patent grants and invention patent grants as robustness checks.[5] As shown in Appendix Table A11, the results remain robust when using grant data instead of application data.

Finally, we conduct a placebo test by randomly assigning treatment groups and years in which firms are included in the entity list. We then estimate Eq. (1) using these false treatment groups, and repeat the process 500 times to generate a distribution of coefficients. The kernel density estimations are shown in Appendix Figs. A2 and A3. The coefficients of the false treatment groups are centered around 0, whereas the true treatment group's coefficients lie significantly to the right of the distribution. This indicates that the estimated treatment effect is not driven by random variation or other unobservable factors.

The above findings increase the validity and reliability of our baseline results, confirming the significantly positive impact of U.S export controls on the innovation performance of Chinese firms in business groups.

[4] Referring to Anwar et al. (2024), we choose four types of covariates to address selection bias caused by differences among firms in factors such as fundamental financial conditions, technological level, overseas operations, and political connections that make one particular type of firms more likely to be included in the entity list. Further, the results of the first stage and the balance test of cross-sectional PSM are shown in Appendix Table A7, Table A8a, Table A8b, Fig. A1a and Fig. A1b.

[5] To address the truncation problem of invention patent grants, we do the following: first, we count the time difference between invention patent application and granting of all listed companies from 1990 to 2017, and get the average distribution from application to granting W_s, i.e., the average proportion of patents granted in the sth year of application. Since invention patents are generally granted with a lag of 0–4 years, invention grants for 2018–2022 are likely to be underestimated in our data. We adjusted the invention patent grants for 2018–2022 with W_s, using the specific adjustment formula: $P_{adj} = \frac{P_{raw}}{\sum_{s=0}^{2022-t} W_s}$, where P_{adj} is the adjusted invention patent grants, P_{raw} is invention patents applied for in year t and granted by the end of 2022. This gave us the adjusted invention patent grants (*lngrant_inv*). Next, since utility and design patent grants generally does not have a lag problem, we counted grants of these two categories and summed them up with the invention grants to get the adjusted patent grant data of listed companies (*lngrant*).

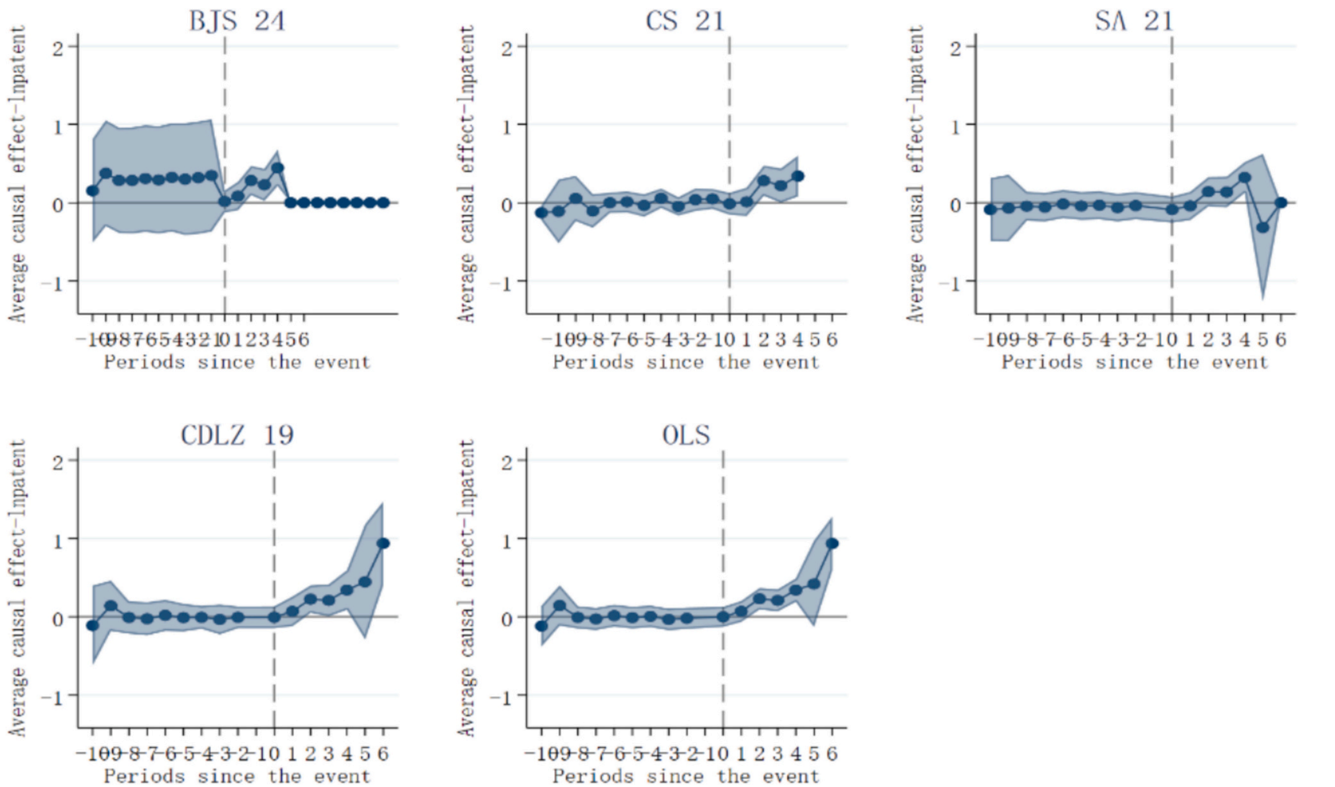

Fig. 4. Estimates of various robustness methods. Notes: This figure presents the results with the dependent variable being *lnpatent*, and the results are similar when the dependent variable is *lninvent*, which is omitted here. BJS 24, CS 21, SA 21, CDLZ 19 are the estimation methods proposed by Borusyak et al. (2024), Callaway and Sant'Anna (2021), Sun and Abraham (2021), and Cengiz et al. (2019), respectively.

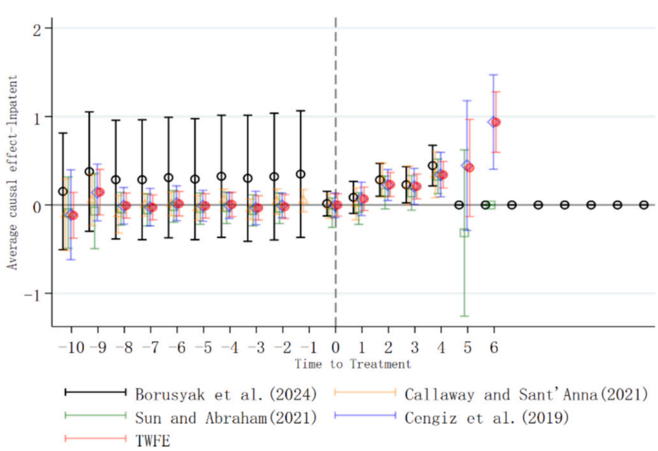

Fig. 5. Estimates of various robustness methods. Notes: This figure presents the results with the dependent variable being *lnpatent*, and the results are similar when the dependent variable is *lninvent*, which is omitted here. The black, orange, green, blue, and red lines represent results from Borusyak et al. (2024), Callaway and Sant'Anna (2021), Sun and Abraham (2021), Cengiz et al. (2019) and two-way fixed effects, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

6. Mechanisms

The baseline results presented in Section 5 suggest that U.S. export controls serve as a good catalyst for Chinese firms' innovation in business groups. In this section, we interpret the results through two potential mechanisms: internal innovation transfer and overall innovation enhancement, and identify which one is more consistent with the results.

6.1. Internal innovation transfer

If business groups function as internal knowledge and technology markets, the interactions among firms within the business groups have important implications for individual firms. Given that industrial interactions between firms in business groups are closely related to their innovation activities, we divide the indirectly shocked firms into same-industry and different-industry groups according to whether they are in the same industry as the blacklisted firms. The regression results are shown in Panel A and Panel B of Table 5, respectively.

The results in Panel A of Table 5 show that indirectly shocked firms in the same-industry groups experience a significant increase in invention patent applications compared to unaffected firms after the sanctions, although there is no increase in patent applications or innovation efficiency.[6] To further investigate the reasons behind the increase in invention patents and verify the hypothesis of peer innovation substitution, we use technology mergers and acquisitions (*techma*) and invention patent transactions (*invent_transactions*) as the dependent variables in Eq. (1). Mergers and acquisitions data for listed firms are obtained from the CSMAR database. Following Ahuja and Katila (2001), we define a firm's mergers and acquisitions as technology-related if it meets any of the following criteria: 1) the merger or acquisition mentions technology or patents as part of the asset transfer, 2) the acquired company operates in a high-technology industry and has been involved in patenting activity within the last five years before the acquisition.

[6] We find a significant decline in utility patent applications and a decline in design patent applications, suggesting that the shock had a positive impact mainly on the application of high-quality patents (invention patents) by firms, but not on the application of low-quality patents (utility and design patents). For the sake of simplicity, we have omitted the results for utility and design patents, which have been retained.

Table 5
Internal innovation transfer.

VARIABLES	(1) *lnpatent*	(2) *lninvent*	(3) *lnpatent*	(4) *lninvent*	(5) *efficiency*	(6) *efficiency_inv*
Panel A: Same-industry groups						
entitylist	−0.0000	0.1939**	−0.0261	0.1792**	−0.0002	0.0000
	(0.093)	(0.086)	(0.094)	(0.086)	(0.000)	(0.000)
Control Vars.	No	No	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8585	8585	8585	8585	7110	7110
R-squared	0.830	0.818	0.831	0.819	0.494	0.501
Panel B: different-industry groups						
entitylist	0.1019**	0.1369***	0.0961**	0.1360***	0.0011**	0.0004**
	(0.047)	(0.041)	(0.047)	(0.041)	(0.000)	(0.000)
Control Vars.	No	No	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,071	10,071	10,071	10,071	8374	8374
R-squared	0.843	0.834	0.844	0.835	0.485	0.467
Panel C: supply-chain groups						
entitylist	0.1305**	0.1574***	0.1306***	0.1610***	0.0016***	0.0004**
	(0.051)	(0.045)	(0.051)	(0.045)	(0.001)	(0.000)
Control Vars.	No	No	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9651	9651	9651	9651	8016	8016
R-squared	0.845	0.835	0.846	0.836	0.485	0.473
Panel D: non-supply chain groups						
entitylist	0.1135**	0.1089**	0.1081**	0.1086**	0.0011**	0.0003
	(0.052)	(0.045)	(0.051)	(0.045)	(0.001)	(0.000)
Control Vars.	No	No	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9660	9660	9660	9660	7996	7996
R-squared	0.840	0.831	0.841	0.832	0.475	0.469

Notes: Table 5 presents the results of the internal innovation transfer mechanism. To explore whether the mechanism holds, we divide indirectly shocked firms into same-industry and different-industry (Panel A and Panel B) groups according to whether they are in the same industry as the blacklisted firms, and further we classify the different-industry groups into supply-chain (B is in A's upstream industry) and non-supply chain groups (Panel C and Panel D). All variables are defined in Table 1. All regressions control for firm-level control variables and include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Otherwise, it is classified as a non-technology merger or acquisition. We also track the number of patents and invention patents sold to companies during the sample period. The original data on invention patent transactions are sourced from the incoPat database. By cleaning and parsing the transferee names and matching them with the names of listed firms, we calculate the total number of invention patents sold to listed firms during the sample period. The results, shown in Table 6, demonstrate that *techma* has not significantly increased, while *invent_transactions* has increased significantly. It indicates that the increase in invention patent applications observed in Panel A of Table 5 is driven by invention patent transactions between firms. These findings support the notion of peer innovation substitution, where indirectly affected peer firms actively engage in patenting activities in place of the blacklisted firms. It provides support for H1.

The results in Panel B of Table 5 reveal that indirectly shocked firms in different industries, apart from the blacklisted firms, experience a significant increase in patent applications, invention patent applications, and innovation efficiency compared to unshocked firms after the sanctions, which is consistent with the baseline results. To further investigate the underlying mechanisms of how firms' innovation in

Table 6
Mechanism analysis of same-industry groups.

VARIABLES	(1) *techma*	(2) *invent_transactions*
entitylist	−0.0067	2.3709**
	(0.018)	(1.153)
Control Vars.	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Observations	8585	8585
R-squared	0.323	0.642

Notes: Table 6 shows the mechanism analysis of same-industry groups. *techma* equals 1 if the underlying details of the merger and acquisition mentions technology or patents as part of the asset transfer, or the acquired company operates in a high-technology industry and has engaged in patenting activity within the last five years before the acquisition. *invent_transactions* represents the number of invention patents sold to companies. All main variables are defined in Table 1. All regressions control for firm-level control variables and include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

different industries respond to indirect shocks, we take two additional steps. First, we classify the different-industry relationships into supply-chain (where firm B operates in the upstream industry of firm A[7]) and non-supply chain groups. The results, presented in Panel C and Panel D, show that the coefficient estimates in Panel C are larger and more significant, suggesting that supply chain relationships are of great importance to firms' innovation. Second, we calculate the proportion of firms' patents citing U.S. patents, the proportion of firms' invention patents citing U.S. patents, the propensity for Chinese patents to cite U.S. patents relative to citing Chinese ones, and the propensity for Chinese invention patents to cite U.S. patents relative to citing Chinese ones, as new dependent variables and run regressions within supply-chain groups. The latter two variables are constructed referring to Han et al. (2024) as follows:

$$\mathrm{p}_{\mathrm{c,u,t}} = \frac{\frac{\mathrm{n}_{\mathrm{c,u,t}}}{\mathrm{x}_{\mathrm{u,t}}}}{\frac{\mathrm{n}_{\mathrm{c,c,t}}}{\mathrm{x}_{\mathrm{c,t}}}} \tag{3}$$

where $n_{c,u,t}$ ($n_{c,c,t}$) is the number of citations made by Chinese patents or invention patents to U.S. patents (Chinese patents) in year t, and $x_{u,t}$ ($x_{c,t}$) denotes the total number of patents granted by U.S. (Chinese) national patent offices in year t, which are used to normalize the citation numbers. We use these four variables to measure China's technological dependence on and decoupling from the United States. The results, presented in Table 7, show that in supply-chain groups, export controls on downstream firms prompt upstream firms to innovate independently. This, in turn, accelerates the technological decoupling of upstream firms from the United States, verifying the hypothesis of supply chain-driven innovation progress (H2).

The above results support the internal innovation transfer hypothesis, indicating the existence of an internal knowledge and technology market within business groups. They demonstrate that export controls drive the reallocation of innovation resources from blacklisted firms to other firms, especially upstream firms. To further explore the channels through which these transfers occur, we examine the reallocation of finance and talent within the group, with the results presented in Table 8. In the first two columns, the dependent variables are the KZ index (Kaplan and Zingales, 1997) and cash holdings (Opler et al., 1999), which measure the extent of firms' financing constraints.[8] The coefficient estimates suggest that, following the shock, firms experience reduced financial constraints and hold less cash, which is consistent with previous literature on the loosening of financial constraints (Almeida et al., 2004; Han and Qiu, 2007). The last four columns focus on human capital structure, using the percentage of employees with graduate degrees, bachelor's degrees, specialist qualifications, and high school or below education to measure a firm's talent composition. The results show that, following the shock, firm' human capital structures reorganize, with an increase in the share of higher-educated employees (graduates and undergraduates) and a decrease in the share of lower-educated employees (specialists and those with high school education or below).

Overall, the results in Table 8 indicate that both capital and talent are reallocated within the business groups following the shock, further validating the internal innovation transfer hypothesis. It suggests that the specific transfer channels involve the reallocation of capital and talent. Taken together, these results support H3 and H4.

Table 7
Mechanism analysis of supply-chain groups.

	(1)	(2)	(3)	(4)
VARIABLES	*ratio*	*ratio_inv*	*decoupling*	*decoupling_inv*
entitylist	−0.0516**	−0.0526**	−0.3758**	−0.0711**
	(0.022)	(0.023)	(0.161)	(0.031)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	3389	3316	3388	3314
R-squared	0.365	0.356	0.620	0.591

Notes: Table 7 shows the mechanism analysis of supply-chain groups. *ratio* represents the proportion of firms' patents citing U.S. patents, and *ratio_inv* represents the proportion of firms' invention patents citing U.S. patents. *decoupling* represents the propensity for Chinese patents to cite U.S. patents relative to citing Chinese ones and *decoupling_inv* represents the propensity for Chinese invention patents to cite U.S. patents relative to citing Chinese ones. All main variables are defined in Table 1. All regressions control for firm-level control variables and include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

6.2. Overall innovation enhancement

If the U.S. entity list leads to an innovation boost for the entire business group, we would expect to observe better innovation performance for indirectly shocked firms as well. Therefore, it is essential to consider the impact of the shock on the business group as a whole. For this reason, we treat groups with at least one firm affected by the U.S. entity list as the treatment group, and groups with all firms unaffected by the shock as the control group. We then aggregate the sample to the group level, re-run the baseline regressions and test the effect of export controls on the overall innovation of the group. Columns (1)–(2) in Table 9 control for group-level fixed effects and year fixed effects, and columns (3)–(6) control for the same control variables at the group level as in the baseline regression, in addition to controlling for group-level and year fixed effects. The results show that the coefficients are insignificant, regardless of whether the dependent variables are patent applications, invention patent applications, or innovation efficiency. It suggests that the innovation performance of the group as a whole is not significantly improved. The results support the internal innovation transfer hypothesis while rejecting the overall innovation enhancement hypothesis (H5).

6.3. Further analyses

The above results demonstrate that export controls have a positive effect on the innovation performance of firms within a business group, and that intra-group innovation transfers occur in response. To further explore the broader effects, we test the impact of the U.S. entity list on the productivity of firms in business groups. We calculate a firm's total factor productivity (TFP) using OP method, LP method, and GMM method, respectively (Olley & Pakes, 1996; Levinsohn and Petrin, 2003; Blundell and Bond, 2000). The results in Table 10 shows that export controls significantly improve firms' productivity, illustrating positive spillovers of export controls on product markets as well.

7. Conclusions

This paper investigates the impact of U.S. export controls on the innovation performance of firms in Chinese business groups. We employ a multiple-period difference-in-differences approach to examine how the U.S. entity list affects firms that are indirectly shocked in business

[7] We determine the upstream industries based on the sectors of a listed company's top five suppliers.

[8] The KZ index can be estimated by the equation $KZindex = \alpha_1 \frac{CF_t}{A_t} + \alpha_2 \frac{Div_t}{A_t} + \alpha_3 \frac{Cash_t}{A_t} + \alpha_4 Leverage_t + \alpha_5 Q_t$, where A_t is the total asset, CF_t is the operating net cash flow, Div_t is the dividend, $cash_t$ is the cash and cash equivalents, $Leverage_t$ is the debt to asset ratio, and Q_t is the Tobin's Q value. Detailed algorithm for calculating the KZ index can be found in Kaplan and Zingales (1997). Cash holdings are calculated as $cashhold = cash\ and\ cash\ equivalents / (total\ assets - cash\ and\ cash\ equivalents)$.

Table 8
Governance channels: reallocation of capital and talents.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	*KZindex*	*cashhold*	*graduate*	*bachelor's*	*specialties*	*high school and below*
entitylist	−0.0858**	−0.0388***	0.8282***	0.8233**	−1.7791***	−1.0496**
	(0.043)	(0.008)	(0.138)	(0.324)	(0.278)	(0.519)
Control Vars.	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,548	10,989	7910	9275	8622	6029
R-squared	0.844	0.691	0.892	0.879	0.794	0.848

Notes: Table 8 presents the governance channels of business groups, the reallocation of capital and talent. All main variables are defined in Table 1. All regressions control for firm-level control variables and include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 9
Overall innovation enhancement.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	*lnpatent*	*lninvent*	*lnpatent*	*lninvent*	*efficiency*	*efficiency_inv*
entitylist	0.2688	0.2866	0.2845	0.2988	0.0017	0.0004
	(0.270)	(0.254)	(0.265)	(0.248)	(0.003)	(0.002)
Control Vars.	No	No	Yes	Yes	Yes	Yes
Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2257	2257	2257	2257	2194	2194
R-squared	0.845	0.845	0.851	0.851	0.741	0.759

Notes: Table 9 shows the results of the overall innovation enhancement mechanism. To test the effect of the U.S. entity list on the overall innovation performance of the group, we aggregate the sample to the group level. All variables are defined in Table 1. All regressions control for group-level control variables and include group fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 10
Further analyses.

	(1)	(2)	(3)
VARIABLES	*TFP_OP*	*TFP_LP*	*TFP_GMM*
entitylist	0.0464***	0.0418***	0.0514***
	(0.015)	(0.015)	(0.019)
Control Vars.	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	10,055	10,055	10,055
R-squared	0.922	0.936	0.867

Notes: Table 10 presents the results of the further analyses. *TFP_OP*, *TPF_LP* and *TFP_GMM* represent firms' TFP using the OP method, the LP method, and the GMM method, respectively. All variables are defined in Table 1. All regressions control for firm-level control variables and include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

groups. Our analysis yields three main conclusions. First, the U.S. entity list has a positive effect on firms' innovation performance. The result remains valid after a series of robustness tests. Second, the positive effect of export controls persists when the indirectly shocked firms are peers of the blacklisted firms, and the effect is more pronounced when the indirectly shocked firms are upstream of the blacklisted firms. After the sanctions, both capital and talent are reallocated within the group to the indirectly shocked firms. These findings support the internal innovation transfer hypothesis, which posits that business groups serve as an internal innovation market, with innovation transfers occurring through inter-firm industrial interactions and resource reallocation. In other words, innovation transfer within the group is facilitated by peer competition-induced innovation substitution and supply chain-driven innovation advancement, resulting in the reallocation of capital and talent to the indirectly shocked firms. Furthermore, we exclude the overall innovation enhancement hypothesis by testing that shocks do not have a significant effect on the innovation performance of the group as a whole. Third, our empirical results show that export controls significantly improve firms' productivity, highlighting positive spillovers in product markets as well.

Our findings have important theoretical and practical implications. Theoretically, we validate that a business group can act as an internal innovation market that reallocates innovation factors such as capital and talent in response to external shocks. Our study contributes to the literature on business groups by providing new evidence on the role of the pyramidal business structures in developing countries' business organizations. Empirically, we uncover an unintended innovation-promoting effect of export controls: changes in demand from downstream firms in the supply chain stimulate technological innovation in upstream firms, leading to a certain degree of technological decoupling from developed countries that impose sanctions.

CRediT authorship contribution statement

Yuhan He: Writing – review & editing, Formal analysis, Methodology, Writing – original draft. **Jinqiu Lyu:** Methodology, Writing – original draft, Conceptualization, Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1
Control for industry-year fixed effects.

	(1)	(2)	(3)	(4)
VARIABLES	*lnpatent*	*lninvent*	*efficiency*	*efficiency_inv*
entitylist	0.0583	0.1334***	0.0014***	0.0006***
	(0.047)	(0.041)	(0.001)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes
Observations	10,866	10,866	9123	9123
R-squared	0.849	0.840	0.514	0.513

Notes: All variables are defined in Table 1. All regressions control for firm-level control variables. And we also include firm and industry-year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table A2
Control for both city-year and industry-year fixed effects.

	(1)	(2)	(3)	(4)
VARIABLES	*lnpatent*	*lninvent*	*efficiency*	*efficiency_inv*
entitylist	0.0657	0.1670***	0.0011**	0.0005**
	(0.056)	(0.050)	(0.000)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes
Observations	9630	9630	7901	7901
R-squared	0.878	0.871	0.622	0.625

Notes: All variables are defined in Table 1. All regressions control for firm-level control variables. And we also include firm fixed effects, city-year and industry-year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table A3
Zero-inflated Poisson regression and zero-inflated Negative Binomial regression.

	(1)	(2)	(3)	(4)
	Zero-inflated Poisson		Zero-inflated negative binomial	
VARIABLES	*patent*	*invent*	*patent*	*invent*
entitylist	1.3081***	1.5293***	1.3058***	1.5420***
	(0.005)	(0.006)	(0.069)	(0.086)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	11,131	11,131	11,131	11,131
Vuong test	33.96***	39.39***	82.89***	82.27***
lnalpha	/	/	0.8476***	0.8709***

Notes: *patent* and *invent* are raw patent applications and invention patent applications, respectively. All variables are defined in Table 1. All regressions control for firm-level control variables and include firm and year fixed effects. We use zero-inflated Poisson and zero-inflated Negative Binomial methods, respectively. Columns 1 and 2 are the results of the zero-inflated Poisson regressions, and Columns 3 and 4 are the results of the zero-inflated Negative Binomial regressions. All coefficients reported here are incidence rate ratios. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table A4
The inverse hyperbolic sine transformation.

	(1)	(2)
VARIABLES	*lnpatent_hat*	*lninvent_hat*
entitylist	0.1318***	0.2247***
	(0.050)	(0.045)
Control Vars.	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Observations	10,989	10,989
R-squared	0.823	0.814

Notes: *lnpatent_hat* and *lninvent_hat* are patent applications and invention patent applications after the inverse hyperbolic sine transformation. All variables are defined in Table 1. All regressions control for firm-level control variables and include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table A5
Regression results after removing samples with changes in shareholdings.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	*lnpatent*	*lninvent*	*lnpatent*	*lninvent*	*efficiency*	*efficiency_inv*
entitylist	0.1400***	0.2216***	0.1238***	0.2124***	0.0011**	0.0005***
	(0.044)	(0.038)	(0.044)	(0.038)	(0.000)	(0.000)
Control Vars.	No	No	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,814	10,814	10,814	10,814	9108	9108
R-squared	0.830	0.822	0.831	0.823	0.451	0.450

Notes: All variables are defined in Table 1. All regressions control for firm-level control variables and include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table A6
PSM-DID (nearest neighbor matching for each year).

	(1)	(2)	(3)	(4)
VARIABLES	*lnpatent*	*lninvent*	*efficiency*	*efficiency_inv*
entitylist	0.1180***	0.2015***	0.0010**	0.0004***
	(0.044)	(0.038)	(0.000)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	10,713	10,713	9008	9008
R-squared	0.828	0.818	0.451	0.455

Notes: All variables are defined in Table 1. All regressions control for firm-level control variables and include firm and year fixed effects. We use nearest neighbor PSM for each year. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table A7
The results of the first stage.

	(1)	(2)	(3)	(4)
VARIABLES	*entitylist*	*entitylist*	*entitylist*	*entitylist*
	Logit	Logit	Logit	Logit
size	0.124***	0.124***	0.121***	0.121***
	(0.025)	(0.025)	(0.027)	(0.027)
age	−0.142***	−0.142***	−0.021	−0.021
	(0.038)	(0.038)	(0.040)	(0.040)
lev	−0.698***	−0.698***	−0.549***	−0.549***
	(0.146)	(0.146)	(0.156)	(0.156)
cash	−0.390*	−0.390*	−0.283	−0.283
	(0.204)	(0.204)	(0.219)	(0.219)
roa	−2.635***	−2.635***	−2.491***	−2.491***
	(0.487)	(0.487)	(0.508)	(0.508)
growth	0.031*	0.031*	0.149***	0.149***

(*continued on next page*)

Table A7 (*continued*)

	(1)	(2)	(3)	(4)
VARIABLES	*entitylist*	*entitylist*	*entitylist*	*entitylist*
	(0.016)	(0.016)	(0.021)	(0.021)
dual	−0.736***	−0.736***	−0.716***	−0.716***
	(0.081)	(0.081)	(0.085)	(0.085)
top1	0.014***	0.014***	0.016***	0.016***
	(0.002)	(0.002)	(0.002)	(0.002)
indep	0.635*	0.635*	0.947***	0.947***
	(0.337)	(0.337)	(0.360)	(0.360)
TFP_LP	0.091**	0.091**	0.031	0.031
	(0.033)	(0.033)	(0.036)	(0.036)
oversea	0.079*	0.079*	−0.027	−0.027
	(0.046)	(0.046)	(0.048)	(0.048)
PC	−0.412***	−0.412***	−0.407***	−0.407***
	(0.052)	(0.052)	(0.057)	(0.057)
Constant	−4.203***	−4.203***	−4.076***	−4.076***
	(0.417)	(0.417)	(0.442)	(0.442)
Observations	10,198	10,198	8685	8685

Notes: This table presents the outcomes of the first stage of the PSM with the outcome variables being *lnpatent*, *lninvent*, *efficiency*, and *efficiency_inv*, respectively. Since there are missing values in *efficiency* and *efficiency_inv*, the number of observations in Columns 3 and 4 are smaller.

Table A8a
The balance check of PSM.

	Unmatched	Mean			%reduct	*t*-test	
VARIABLES	Matched	Treated	Control	%bias	|bias|	t	p > |t|
size	U	23.185	22.811	24.8		12.07	0.000***
	M	23.170	23.126	2.9	88.3	1.14	0.253
age	U	2.614	2.649	−5.9		−2.73	0.006***
	M	2.616	2.608	1.4	75.6	0.55	0.579
lev	U	0.513	0.502	5.4		2.54	0.011**
	M	0.513	0.506	3.4	37.2	1.36	0.174
cash	U	0.171	0.180	−6.9		−3.27	0.001***
	M	0.172	0.172	−0.0	99.9	−0.00	0.998
roa	U	0.030	0.032	−3.3		−1.52	0.128
	M	0.030	0.030	0.1	98.3	0.02	0.982
growth	U	0.467	0.438	2.2		1.02	0.307
	M	0.465	0.481	−1.2	46.4	−0.45	0.650
top1	U	42.012	37.698	28.9		13.48	0.000***
	M	41.887	42.07	−1.2	95.8	−0.47	0.641
indep	U	0.316	0.313	5.5		2.64	0.008***
	M	0.316	0.314	2.7	51.7	1.05	0.292
TFP_LP	U	9.037	8.790	21.6		10.07	0.000***
	M	9.030	9.001	2.5	88.5	0.99	0.320

Notes: This table presents the results of the balance check of the PSM with the outcome variables being *lnpatent* and *lninvent*.

Table A8b
The balance check of PSM.

	Unmatched	Mean			%reduct	t-test	
VARIABLES	Matched	Treated	Control	%bias	|bias|	t	p > |t|
size	U	23.178	22.858	21.3		9.70	0.000***
	M	23.177	23.131	3.0	85.9	1.15	0.250
age	U	2.620	2.617	0.6		0.26	0.793
	M	2.620	2.622	−0.4	27.4	−0.17	0.866
lev	U	0.512	0.497	7.3		3.22	0.001***
	M	0.511	0.508	1.6	77.5	0.64	0.525
cash	U	0.172	0.180	−6.8		−3.06	0.002***
	M	0.172	0.173	−0.9	87.6	−0.33	0.739
roa	U	0.030	0.033	−5.9		−2.56	0.011**
	M	0.030	0.030	−0.3	95.1	−0.11	0.909
growth	U	0.482	0.307	15.8		7.19	0.000***
	M	0.476	0.442	3.1	80.5	1.04	0.298
top1	U	41.924	37.382	30.6		13.46	0.000***
	M	41.901	42.100	−1.3	95.6	−0.50	0.616
indep	U	0.316	0.311	6.7		3.02	0.003***
	M	0.316	0.315	0.5	92.7	0.19	0.853
TFP_LP	U	9.040	8.856	16.4		7.25	0.000***
	M	9.039	9.003	3.2	80.8	1.23	0.219

Notes: This table presents the results of the balance check of the PSM with the outcome variables being *efficiency* and *efficiency_inv*.

Table A9
Control for high-tech enterprise recognition.

VARIABLES	(1) *lnpatent*	(2) *lninvent*	(3) *efficiency*	(4) *efficiency_inv*
entitylist	0.1169***	0.1965***	0.0010**	0.0004***
	(0.043)	(0.038)	(0.000)	(0.000)
qualify	0.3199***	0.2008***	0.0005	0.0000
	(0.049)	(0.043)	(0.000)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	10,989	10,989	9273	9273
R-squared	0.830	0.821	0.450	0.452

Notes: *qualify* is a dummy variable representing whether a company is selected as a high-tech enterprise in that year or not. All regressions control for firm-level control variables and include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table A10
Control for additional deduction from R&D expense.

VARIABLES	(1) *lnpatent*	(2) *lninvent*	(3) *efficiency*	(4) *efficiency_inv*
entitylist	0.0954**	0.1738***	0.0011**	0.0005***
	(0.043)	(0.038)	(0.000)	(0.000)
adpolicy	0.2550***	0.2708***	−0.0011***	−0.0006***
	(0.033)	(0.026)	(0.000)	(0.000)
Control Vars.	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	10,989	10,989	9273	9273
R-squared	0.830	0.821	0.450	0.452

Notes: *adpolicy* is the policy shock term, where 2016 as the year of policy implementation, enterprises in the six negative list industries as the control group, and the remaining industries as the treatment group[a]. All regressions control for firm-level control variables and include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

[a] Six negative list industries are tobacco manufacturing, accommodation and catering, wholesale and retail, real estate, leasing and business services, and entertainment.

Table A11
Results based on granted patents and invention patents.

VARIABLES	(1) *lngrant*	(2) *lngrant_inv*
entitylist	0.1610***	0.2014***
	(0.055)	(0.059)
Control Vars.	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Observations	10,989	10,989
R-squared	0.774	0.670

Notes: All regressions control for firm-level control variables and include firm and year fixed effects. Robust standard errors are reported in the parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

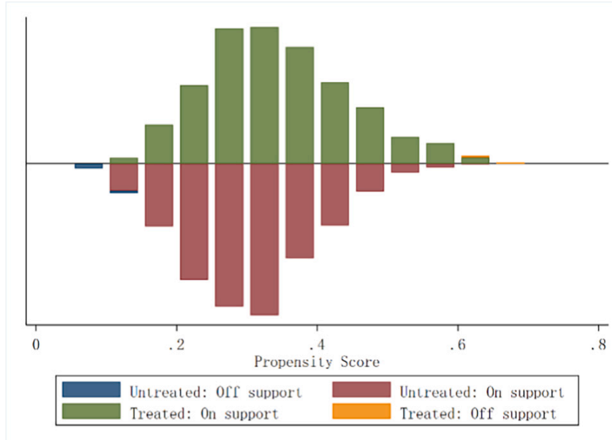

Fig. A1a. On/off support samples of PSM. Notes: This figure presents the results of on/off support samples of the PSM with the outcome variables being *lnpatent* and *lninvent*.

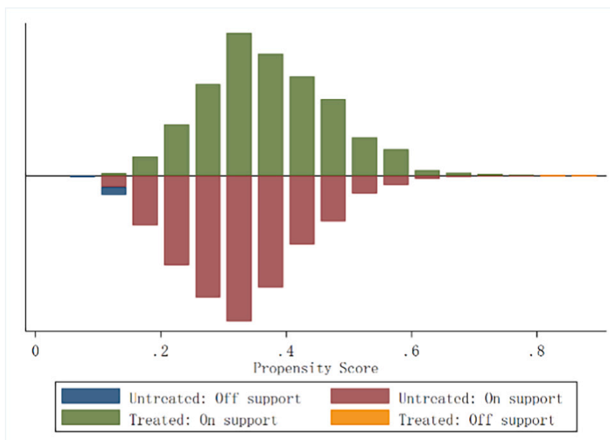

Fig. A1b. On/off support samples of PSM. Notes: This figure presents the results of on/off support samples of the PSM with the outcome variables being *efficiency* and *efficiency_inv*.

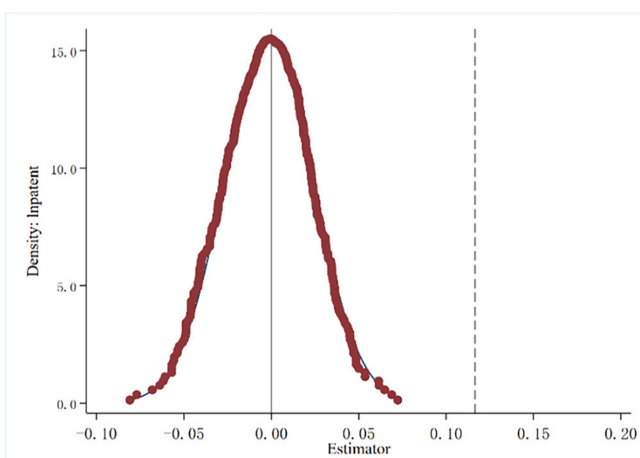

Fig. A2. Placebo test for *lnpatent*. Notes: The red scatter represents the density distribution of the coefficients for patents in the false treatment groups. The gray dashed line represents the magnitude of the coefficients for patents in the true treatment groups. The coefficients of the false treatment groups are centered around 0, whereas the true treatment group's coefficients lie significantly to the right of the distribution, indicating that the estimated treatment effect is not driven by random variation or other unobservable factors.

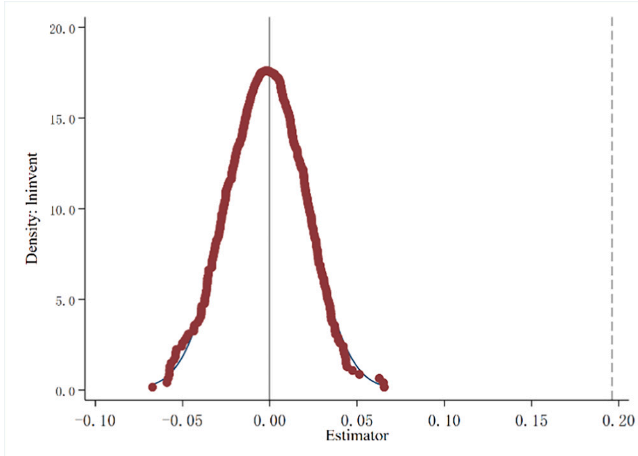

Fig. A3. Placebo test for *lninvent*. Notes: The red scatter represents the density distribution of the coefficients for invention patents in the false treatment groups. The gray dashed line represents the magnitude of the coefficients for invention patents in the true treatment groups. The coefficients of the false treatment groups are centered around 0, whereas the true treatment group's coefficients lie significantly to the right of the distribution, indicating that the estimated treatment effect is not driven by random variation or other unobservable factors.

Data availability

Data will be made available on request.

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